

Arman Chhetri

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Advisor: Yinxi Liu (yxlics@rit.edu)

Research Interests

LLM security (inference-cost / resource-exhaustion stressors; stress-testing LLM inference infrastructure) and software security analysis, including scalable constant-time / side-channel verification.

Education

Ph.D. in Cybersecurity

Aug 2025 – Present

Rochester Institute of Technology (RIT), Rochester, NY

Advisor: Yinxi Liu.

B.E. in Computer Engineering

Nov 2017 – Apr 2022

Pulchowk Campus, IOE, Tribhuvan University, Lalitpur, Nepal

First Division (78.21%). Entrance exam rank 29 / ~18,000 (2017). Capstone: Speech Recognition in Nepali (supervisor: Subarna Shakya, Ph.D.).

High School, Liverpool International College, Kathmandu (Jun 2015 – Jun 2017); aggregate 85.8%, top 5 of class.

Publications

- [1] Arman Chhetri, Yinxi Liu. “Metamorphic Stress Testing of Vision-Language Components for Time-Critical Autonomous Systems.” *TRUST '26*, co-located with ASE 2026, Munich, Germany, Oct 12–16, 2026. Accepted 2026-08-23. DOI: [10.1145/3843782.3844660](https://doi.org/10.1145/3843782.3844660). Zenodo: [10.5281/zenodo.21497533](https://zenodo.org/record/21497533).
- [2] Arman Chhetri, Yinxi Liu. “When Do Inference-Cost Stress Tests Generalize? A Cross-Model and Cross-Configuration Measurement Study.” *RASE '26*, co-located with ASE 2026, Munich, Germany, Oct 12–16, 2026. Accepted 2026-08-25. DOI: [10.1145/3820756.3844910](https://doi.org/10.1145/3820756.3844910). Zenodo: [10.5281/zenodo.21460437](https://zenodo.org/record/21460437).
- [3] M. Dhakal, A. Chhetri, A. K. Gupta, P. Lamichhane, S. Pandey, and S. Shakya. “Automatic speech recognition for the Nepali language using CNN, bidirectional LSTM and ResNet.” *ICICT 2022*, pp. 515–521. DOI: [10.1109/ICICT54344.2022.9850832](https://doi.org/10.1109/ICICT54344.2022.9850832).

Research Experience

Ph.D. Research (Advisor: Yinxi Liu)

Aug 2025 – Present

Rochester Institute of Technology

- LLM inference-cost / resource-exhaustion stressors and metamorphic stress testing of vision-language components for time-critical systems (TRUST '26 / RASE '26).
- Scalable constant-time / side-channel verification **using** CT_Prover: observation-directed relational slicing for cryptographic code (e.g., BearSSL RSA), Boogie/Z3 VCs, SMACK/BAM, and Python CT_Prover drivers. Exploring observation-relevant slice reconstruction to reduce solver cost on RSA paths (in progress; unpublished).

Selected Projects

ContamBench — Dataset and CLIs for contamination stress testing of open-weight LLM serving.

TRUST VLM stress testing — Metamorphic stress testing for vision-language components in time-critical settings.

CT_Prover-based constant-time work — Extending constant-time analysis with observation-directed slicing (not claiming CT_Prover authorship).

ASR for Nepali; Distributed Filestore; HTTP Server — Capstone ASR (CER 17.06%) and from-scratch Go systems projects.

Work Experience

Software Engineer, Logpoint Inc.

Apr 2024 – Jul 2025

Lalitpur, Nepal

- Built and maintained Linux application-server monitoring tools (Go, GraphQL, TypeScript/JS, Python).
- Owned the License Portal lifecycle (TypeScript, Rust, Node.js, GraphQL, gRPC, NestJS, Ansible, GitLab CI, Docker).
- Contributed to AgentX EDR; fixed bugs and performance issues.

Lab Tutor (Part-time), Logpoint Inc. / Tribhuvan University

Jan 2024 – Mar 2024

Lalitpur, Nepal

- Assisted with “Security Operations Fundamentals”; materials, grading, and labs (phishing, threat modeling, SIEM; network/web/Linux/app security via pwn.college).

Solutions Engineer, Logpoint Inc.

May 2023 – Mar 2024

Lalitpur, Nepal

- 150+ L3 cases (OS, application, disk, network); training for new hires; SSL/TLS, PKI, AWS; AgentX bug fixes.

Associate Solutions Engineer, Logpoint Inc.

Apr 2022 – Apr 2023

Lalitpur, Nepal

- SIEM debugging/review; Linux/Windows reliability; crypto, authn/authz, REST, cloud, Nxlog, Wazuh; 450+ cases with RCA across Python, Java, SQL, MongoDB, Docker, Kafka.

Software Developer and Cofounder, Techniti Nepal

Feb 2021 – Mar 2022

Lalitpur, Nepal

- Healthcare and management web apps and a cricket fantasy platform (Python, JavaScript, MySQL, Linux, Docker).

Software Engineering Intern (Part-time), Sireto Technology

Sep 2021 – Dec 2021

Lalitpur, Nepal

- Haskell smart contracts on Cardano; completed Plutus Pioneer Program.

Skills

Languages: Python, Go, Rust, TypeScript/JavaScript, Bash, C; exposure to Haskell.**Research / systems:** LLM inference tooling (vLLM), experimental evaluation / stress testing, static analysis / software security, Linux, Docker.**Selected industry:** GraphQL, gRPC, NestJS, Ansible, CI/CD, SIEM operations, networking and systems troubleshooting.**Certifications (selected)**

ISC2 Certified in Cybersecurity; AWS Academy Graduate; Cisco Networking Essentials; Coursera Algorithmic Toolbox; Coursera Deep Learning Specialization.